

Midterm 2 Review

I have a set of 20 documents and featurize them using `CountVectorizer(max_features=50)`. I calculate the distance between all of the documents using both cosine distance and euclidean distance and get different answers.

What is the difference between the two distances? Which one is preferable and why?

Instead of using CountVectorizer I use TfidfVectorizer.

How does TfidfVectorizer differ from CountVectorizer?

Will I still get a difference in the similarity depending on the distance I use? Why?

Model Identification Speed Round

Identify if the following problems are most suitable for a regression, classification, or clustering model:

A bank monitors transactions in real-time. For every swipe, the system must decide immediately whether the transaction is legitimate or not based off the transaction amount and location. What type of model should they use?

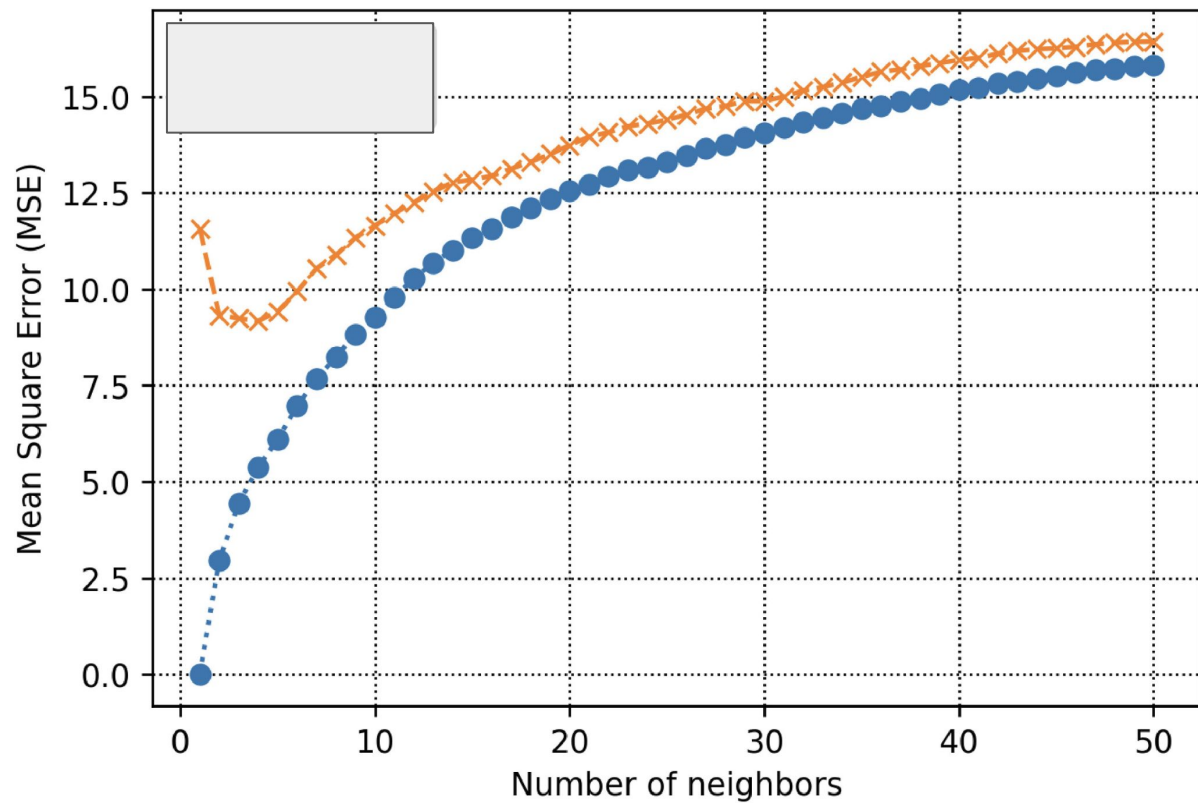
A farmer uses satellite imagery and soil sensor data to estimate the total weight of corn (in bushels) that her field will produce at the end of the season. What type of model should they use?

An executive of a large newspaper wants to get an idea of the different topics the Arts and Entertainment section frequently writes about. She takes all the headlines over the last 5 years and preprocesses them using TfidfVectorizer. What type of model should she use?

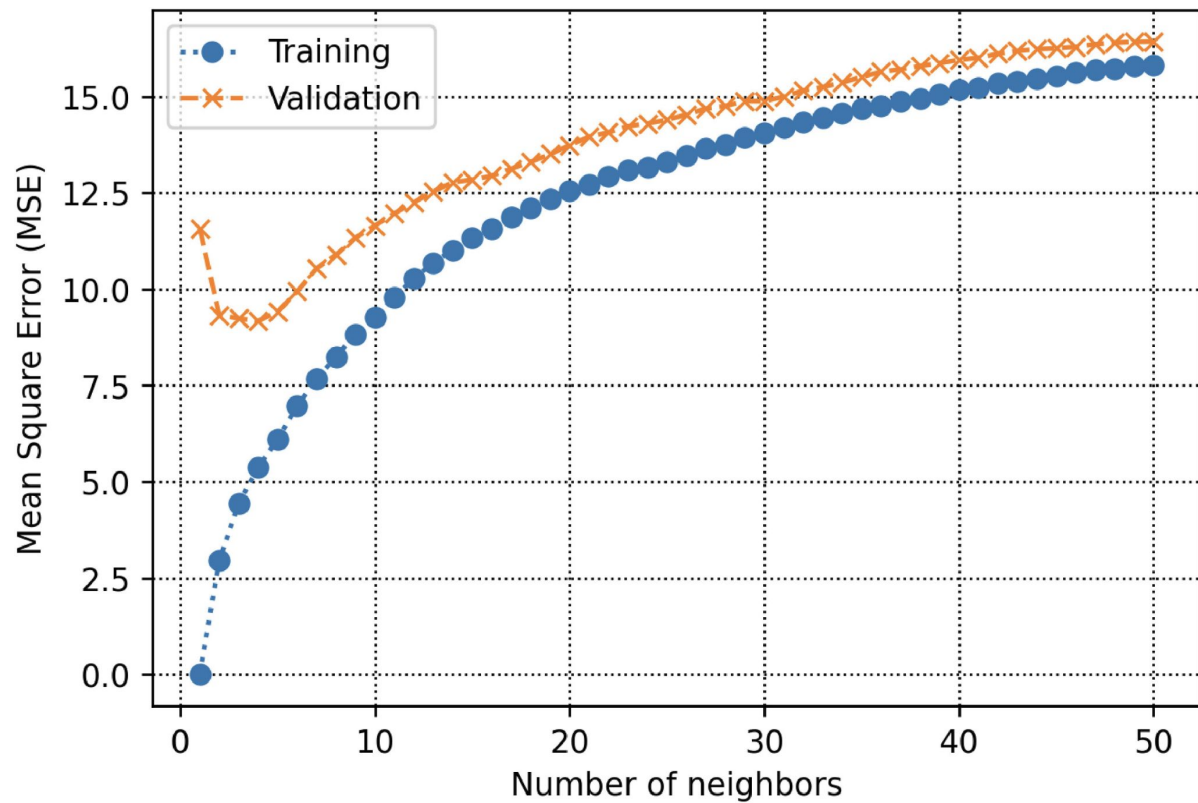
A wearable device tracks a user's heart rate. The device is programmed to alert the user whenever their heart rhythm looks "different" or "strange" compared to their normal baseline. What model should we use?

A self-driving vacuum needs to decide at every second how many degrees to turn its wheels (e.g., 12.5 degrees, 14.2 degrees) to avoid a wall.

Which one is training?



Which one is training MSE?



Identify what's wrong with this ML procedure:

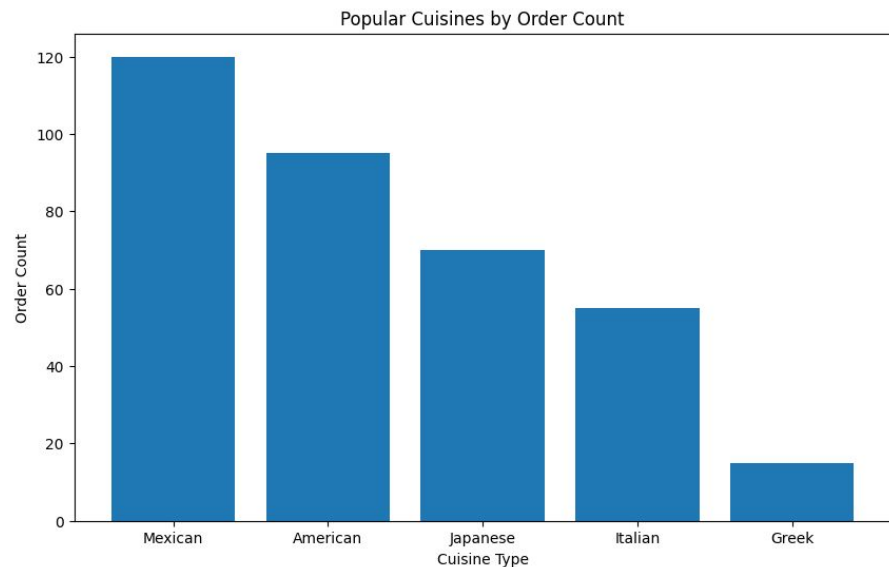
Alex has a dataset where y is 0 or 1 with a **textual** feature, **categorical** feature, and a **quantitative** feature. Their code is below:

```
make_column_transformer(  
    (CountVectorizer(), "text"),  
    (OneHotEncoder(), ["cat"]),  
    remainder=passthrough)  
  
make_pipeline(  
    transformer,  
    KNeighborsRegressor())  
  
pipeline.fit(X, y)  
y_pred = pipeline.predict(X)  
  
root_mean_square_error(y, y_pred)
```

What is the difference between MSE and MAE?

My dataset has a few points that are possible outliers but I don't feel comfortable removing them. I want most predictions to be close, and if they're far away, I don't really care how much. What scoring metric should I use?

Nick has a dataset of different recipes. The y is type of cuisine and the X is tfidf vectorized recipe text. The distribution of the labels are shown below



Nick wants to train a KNeighborsClassifier to predict if a recipe is for Greek food or not. He casts the labels to be 1=Greek 0=Not Greek. He selects the best k using GridSearchCV. His best model has an accuracy of 95.7%. Is this a good model?

BioX is a disease testing company with a very effective test for Disease A but it's expensive and draining for the patient to undergo. However, without a diagnosis and treatment Disease A has a high mortality rate.

They want to train a classification model using biomarkers in the patient's blood to preliminarily screen patients for the disease. What measurement should they use to evaluate their model? Why?

They test 100k patients and get the following confusion matrix (per 1k):

		Predicted	
		-	+
True	-	30	10
	+	20	40

What is the Precision? Recall? F1?

What is the tradeoff between grid search, coordinate optimization, and random search?

I use grid search over k values from 1 to 10 and euclidean distance vs manhattan distance for my pipeline. The grid search finds that I should use $k=10$ and manhattan distance. What should I do?

- a) Save both parameters to my model
- b) Save just the k value
- c) Save just the distance
- d) Save none of the values